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Algebra IV — Normal Exam

Year: 2021

Exercise 1:

Let $a \in \mathbb{R}$. Consider the matrix

$$A_a = \begin{pmatrix} 2 & 2a-10 & -2 \\ -1 & a-1 & -1 \\ 0 & -3a+12 & 4 \end{pmatrix}.$$

1) 1) Determine $\mathrm{Rg}(A_a - I_n)$. Conclusion?
   2) For which values of $a \in \mathbb{R}$ is the matrix A_a diagonalizable?

2) 1) Write the minimal polynomial of A_a.
   2) Let $B = A_a$. Compute B^k for $k \in \mathbb{Z}$.

3) Choose $a = 1$ and set $A = A_1$.
   1) Determine a Jordan reduction of the matrix A.
   2) Let $P \in \mathbb{R}[X]$. Show that $P(A)$ is diagonalizable if and only if $P'(1) = 0$.

Answer Area

Characteristic polynomial and eigenvalues. Compute the characteristic polynomial of A_a (variable λ):

$$\chi_{A_a}(\lambda) = \det(\lambda I - A_a) = (\lambda - a)(\lambda - 4)(\lambda - 1).$$

Hence the eigenvalues of A_a are $a,\ 4,\ 1$ (algebraic multiplicities counted accordingly).

- 1. Rank of $A_a - I$ We have the eigenvalue 1, so $\mathrm{rank}(A_a - I) = 3 - \dim\ker(A_a - I) = 3 -$ (geometric multiplicity of 1). A direct computation shows that for all real a the matrix $A_a - I$ has rank 2 except that this rank does not drop further for any real a (indeed one checks the linear system and finds a one-dimensional eigenspace for 1 for generic a). Thus

  $$\mathrm{Rg}(A_a - I) = 2 \quad \text{(for all real } a\text{)},$$

  so the geometric multiplicity of the eigenvalue 1 is 1. (Concretely: $A_a - I = \begin{pmatrix} 1 & 2a-10 & -2 \\ -1 & a-2 & -1 \\ 0 & -3a+12 & 3 \end{pmatrix}$; one verifies its rows are linearly dependent but not all zero; rank $= 2$.)

2. Diagonalizability If $a \notin \{1, 4\}$ the three eigenvalues $a, 4, 1$ are pairwise distinct, so A_a is diagonalizable over $\mathbb{R}$.

   If $a = 4$, the eigenvalue 4 has algebraic multiplicity 2; computing $\operatorname{rank}(A_4 - 4I) = 1$ shows the geometric multiplicity equals 2, hence A_4 is diagonalizable.

   If $a = 1$, the eigenvalue 1 has algebraic multiplicity 2 but geometric multiplicity 1 (since $\operatorname{rank}(A_1 - I) = 2$), so A_1 is *not* diagonalizable. Therefore A_a is diagonalizable for every real a except $a = 1$.

- 1. Minimal polynomial The minimal polynomial m_{A_a} divides the characteristic polynomial $(X-a)(X-4)(X-1)$ and is determined by sizes of Jordan blocks.
     - If $a \notin \{1, 4\}$ then all eigenvalues are simple and $m_{A_a}(X) = (X - a)(X-4)(X-1)$.
     - If $a = 4$ then 4 is a repeated eigenvalue but A_4 is diagonalizable, so $m_{A_4}(X) = (X-4)(X-1)$.
     - If $a = 1$ then 1 is a repeated eigenvalue and A_1 is not diagonalizable: the Jordan block for 1 has size 2, hence

$$m_{A_1}(X) = (X-1)^2(X-4).$$

  2. Powers B^k ($k \in \mathbb{Z}_{\geq 0}$) We treat cases.
     (a) If A_a is diagonalizable (i.e. $a \neq 1$), write $A_a = PDP^{-1}$ with $D = \operatorname{diag}(a, 4, 1)$. Then for any integer $k \geq 0$,

$$A_a^k = PD^kP^{-1} = P\operatorname{diag}(a^k, 4^k, 1^k)P^{-1} = P\operatorname{diag}(a^k, 4^k, 1)P^{-1}.$$

     (For negative integers k this formula holds provided A_a is invertible; invertibility fails iff one eigenvalue is 0.)

     (b) If $a = 1$ (nondiagonalizable), put $A_1 = P\operatorname{diag}(J_2(1), [4])P^{-1}$ where $J_2(1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Then

$$A_1^k = P\operatorname{diag}(J_2(1)^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}, \qquad k \in \mathbb{N},$$

     so

$$A_1^k = P\begin{pmatrix} 1 & k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4^k \end{pmatrix}P^{-1}.$$

     (If desired one can compute P explicitly and write A_a^k as a polynomial in A_a of degree ≤ 2 using Cayley–Hamilton.)

- 1. Jordan reduction of $A = A_1$ For $a = 1$ we have eigenvalues 1 (algebraic multiplicity 2) and 4 (multiplicity 1) with a single Jordan block of size 2 for the eigenvalue 1. Thus the Jordan normal form is

$$J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix},$$

and there exists an invertible real matrix P such that $A_1 = PJP^{-1}$. (One may compute explicitly a Jordan basis by finding an eigenvector v for 1 and a generalized eigenvector w solving $(A_1 - I)w = v$; for instance any concrete computation gives such a pair and the third vector an eigenvector for 4.)

2. Diagonalizability of $P(A)$ Let $Q(X) \in \mathbb{R}[X]$ (we rename the polynomial to avoid confusion with the change-of-basis matrix). Since A_1 is similar to $\operatorname{diag}(J_2(1), [4])$, we have

$$Q(A_1) \sim \operatorname{diag}(Q(J_2(1)), Q(4)).$$

For the Jordan 2-block one has the classical formula (Taylor expansion / functional calculus for polynomials)

$$Q\big(J_2(1)\big) = \begin{pmatrix} Q(1) & Q'(1) \\ 0 & Q(1) \end{pmatrix}.$$

Thus

$$Q(A_1) \sim \begin{pmatrix} Q(1) & Q'(1) & 0 \\ 0 & Q(1) & 0 \\ 0 & 0 & Q(4) \end{pmatrix}.$$

Hence $Q(A_1)$ is diagonalizable over $\mathbb{R}$ iff the off-diagonal entry $Q'(1)$ vanishes, i.e. iff $Q'(1) = 0$. This proves the claim (with P renamed back to the user's polynomial P).

Exercise 2:

Let A be a nonzero matrix in $M_3(\mathbb{R})$ satisfying $A^3 = -A$.

1) Determine the characteristic polynomial of A.
2) Let $A = M(f, B_3)$. Establish the relation

$$\mathbb{R}^3 = \ker f \oplus \ker(f^2 + \mathrm{id}_{\mathbb{R}^3}).$$

3) * Conclude that the matrix A is similar to

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Answer Area

- Characteristic polynomial Given $A^3 = -A$, we have

$$A^3 + A = 0 \qquad \Longrightarrow \qquad A(A^2 + I) = 0.$$

Hence the minimal polynomial of A divides $X(X^2 + 1)$. Consequently the eigenvalues of A belong to the set $\{0, i, -i\}$. Since A is a real matrix, nonreal eigenvalues occur in complex conjugate pairs. As A is not the zero matrix (hypothesis), it cannot be that all eigenvalues are 0. Therefore the characteristic polynomial is

$$\chi_A(X) = X(X^2 + 1) = X^3 + X.$$

This corresponds to eigenvalues $0, i, -i$ (with multiplicities as indicated).

- Primary decomposition: the direct sum. Write the linear endomorphism as $f : \mathbb{R}^3 \to \mathbb{R}^3$ with matrix A in some basis. Consider the polynomials $p_1(X) = X$ and $p_2(X) = X^2 + 1$. They are coprime in $\mathbb{R}[X]$ and satisfy $p_1(f)p_2(f) = 0$ (since $A(A^2 + I) = 0$). By the primary decomposition (or structure theorem for finitely-generated modules over a PID) we obtain

$$\mathbb{R}^3 = \ker p_1(f) \oplus \ker p_2(f) = \ker f \oplus \ker(f^2 + \mathrm{id}).$$

  Concretely, $\ker f$ is the real eigenspace for 0, and $\ker(f^2 + \mathrm{id})$ is a two-dimensional f-stable subspace on which f acts with minimal polynomial $X^2 + 1$.

- Real canonical form. On the two-dimensional invariant subspace $W := \ker(f^2 + \mathrm{id})$ the restriction $f_{|W}$ satisfies $(f_{|W})^2 = -\mathrm{id}_W$, so in a suitable real basis of W the matrix of $f_{|W}$ is the 2×2 real block corresponding to multiplication by the complex number i, namely

$$R := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

  Together with a basis vector of $\ker f$ (on which f acts by 0), we obtain a basis of $\mathbb{R}^3$ in which the matrix of f is

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix},$$

  which is exactly the matrix claimed. Thus A is similar to that block diagonal form.

  (This is the real rational canonical / real Jordan form for the polynomial factorization $X(X^2 + 1)$.)